Microbots powered by soft actuators can fly like insects

MIT researchers have pioneered a new fabrication technique that enables them to produce low-voltage, power-dense, high-endurance soft actuators for an aerial microrobot (Credit: https://news.mit.edu)

MIT researchers have demonstrated diminutive drones that can zip around with bug-like agility and resilience, which could eventually perform tasks such as pollinating a field of crops or searching for survivors amid the rubble of a collapsed building. The soft actuators that propel these microrobots are very durable and operate with 75% lower voltage than their earlier versions while carrying 80% more payload. The actuators are like artificial muscles that rapidly flap the robot's wings. The artificial muscles with fewer defects dramatically extend the lifespan of the components and increase the robot's performance and payload.

Multi-sensor system for more accurate knee implant fitting

CEA-Leti has developed a smart, multi-sensor system for knee implants, thus allowing surgeons to position the implant with high accuracy, significantly reduce the risk of follow-up surgery, and enhance rehabilitation. Called FollowKnee, the system integrates a deformation sensor, a pH sensor, a temperature sensor, and an accelerometer in a titanium tibial baseplate. A more accurate implant fitting is achieved via the deformation sensor and accelerometer that guide the surgeon during the operation. The pH sensor with living tissue, the system's pH, and temperature sensors detect infection early, while the deformation sensor and accelerometer also trigger an alert if they detect mechanical problems, such as loosening of the implant.

Clear acrylic column and tablet for demonstrating FollowKnee. The green device shows comparative size of the system (Credit: www.leti-cea.com)